TEACH YOURSELF INTERMEDIATE XII UNIT 4

Data and Analysis (Python)

DB connectivity, stats, and visualization

What you will master

- Connect Python to databases
- Compute mean, median, mode, spread
- Create bar charts, histograms, scatter plots, box plots

How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

4.1 Data analysis pipeline

Collect → clean → analyze → visualize → decide. Python libraries (pandas, matplotlib) automate heavy lifting.

4.2 Central tendency & spread

Mean: — . Median — middle value; mode — most frequent.

Range, variance, and standard deviation show spread — essential before comparing classes or sales regions.

4.3 Visualization choices

Bar

Compare categories

Histogram

Distribution of numeric data

Scatter

Relationship between two variables

Box plot

Median, quartiles, outliers

Teach Yourself — Practice & Check

1. Given five test scores, compute mean and median by hand.
2. When would a scatter plot mislead if axes are not labeled?